

Altimeter

The Seaglider's altimeter is an acoustic bottom detector: it pings downward during the dive so the glider can start its apogee maneuver a safe margin off the seafloor instead of trusting `$D_TGT` or a bathymetry map. The same board is also a **transponder/pinger**, usable with ship tracking systems for relocation. Unlike a dedicated downward altimeter, though, it is effectively **omnidirectional** — a fact that drives most of its quirks.

Source

Paraphrased from the APL-UW IOP office-hours material ("How to properly use the altimeter", April 2026), the APL-UW *SGX Documentation*, and the Applied Acoustic Engineering *LF Transponder/Altimeter Board Set Operation Manual* (SPC-5414). Check parameter definitions against the [Parameter Reference Manual](#) for your firmware.

First question: do you actually need it?

IOP's advice starts a step earlier than configuration — the altimeter is not always the right tool:

- It uses **power** that could go to science or endurance.
- **Bathymetry maps are often more consistent:** a good `bathymap` on the compact flash lets the glider choose dive depth without pinging at all, and `$D_OFFGRID` (the depth assumed when off the map) is another option.
- **False positives cost energy:** every phantom bottom means a short dive, and short dives spend proportionally more energy per distance.
- **Missed detections cost more:** a non-responding altimeter on an altimeter-dependent mission ends with the glider on the bottom.
- It is **omnidirectional** — the nearest big reflector *becomes* the range, whether that is the seabed, a wall, or the surface (see the `$ALTIM_PING_DEPTH` notes at [seaglider.pub/parms](#) for how the surface return is ignored).